

central intuitions: *low distortion* and *human context*. HIP operationalizes these intuitions with a detector-agnostic pipeline that minimally adapts a base model into a paraphraser and then applies that paraphraser iteratively. The pipeline has three stages: data preparation, minimal fine-tuning, and iterative paraphrasing. We describe each stage in the following subsections.

3.1 Data Preparation

The first stage constructs paired examples $\mathcal{D} = \{(a_i, h_i)\}_{i=1}^M$, where h_i is a high-quality human passage and a_i is an AI paraphrase of the same passage. Here, the direction of the pair matters: we will ultimately train a model to map from the AI text back to the human text.

As summarized in Algorithm 1, the raw corpus is first narrowed to a candidate set by applying basic corpus filters, for example on provenance, length, or document integrity. These candidates are then normalized into a common textual form and deduplicated at the corpus level. After that, a text-quality screen removes passages that are poor targets for rewriting. Only then do we construct pairs. For each remaining human passage h_i , an external paraphraser generates an AI-style rewrite a_i . Pair construction uses bounded rejection and re-sampling: candidates that fail anomaly checks or semantic-preservation checks are discarded and regenerated, and the example is dropped if no valid paraphrase is obtained within a fixed retry budget. Essentially, HIP constructs and trains on *filtered human targets* and *meaning-preserving AI-style sources*, rather than on arbitrary raw text.

Algorithm 1 Data Preparation

Input: A human corpus $\mathcal{C}_{\text{raw}}$ and the paraphrasing retry budget K

Output: Paired dataset $\mathcal{D} = \{(a_i, h_i)\}$

```

1:  $\mathcal{C}_{\text{cand}} \leftarrow \text{FILTERBYPROVENANCEANDLENGTH}(\mathcal{C}_{\text{raw}})$ 
2:  $\mathcal{C}_{\text{dedup}} \leftarrow \text{DEDUPLICATE}(\text{NORMALIZE}(\mathcal{C}_{\text{cand}}))$ 
3:  $\mathcal{C}_{\text{clean}} \leftarrow \{h \in \mathcal{C}_{\text{dedup}} : \text{TEXTQUALITYOK}(h)\}$ 
4:  $\mathcal{D} \leftarrow \emptyset$ 
5: for passage  $h \in \mathcal{C}_{\text{clean}}$  do
6:   for attempt  $\in \{1, \dots, K\}$  do
7:      $a \leftarrow \text{AIPARAPHRASE}(h)$ 
8:     if  $\text{ANOMALYFREE}(a)$  and  $\text{SEMANTICPRESERVATIONOK}(a, h)$  then
9:        $\mathcal{D} \leftarrow \mathcal{D} \cup \{(a, h)\}$ 
10:    break
11: return  $\mathcal{D}$ 
```

3.2 Minimal Fine-Tuning

Given the paired dataset $\mathcal{D}$, the second stage trains a paraphraser $\mathcal{M}_{\text{para}}$ from a pretrained language model $\mathcal{M}_{\text{base}}$ while perturbing the model as little as possible to preserve *low distortion*. HIP therefore uses *minimal fine-tuning*: we do not train a full assistant. Instead, we apply supervised fine-tuning to $\mathcal{D}$, optionally with a parameter-efficient update such as low-rank adaptation [Hu et al., 2022].

The supervision format is likewise kept simple. Rather than using a chat template, we consider paraphrasing as a plain text continuation problem with lightweight structural tags. For a single pair (a, h) , where a is the AI paraphrase and h is the original human passage, the model sees:

```

<source_text>
  AI paraphrase a
</source_text>

<target_text>
  Original human passage h
</target_text>
```

Operationally, the text between the `<source_text>` tag and the `<target_text>` tag form the prompt prefix, while the original human passage and closing `</target_text>` tag form the completion. Training then uses the standard next-token objective, but the loss is restricted to the completion span only. In other words, the model is optimized to reconstruct the human passage conditioned on the AI paraphrase, not to imitate a conversational interface via a chat template.

3.3 Iterative Paraphrasing

Once the paraphraser is trained, the final stage applies it to transform a machine-like passage $x^{(0)}$ into a rewrite $x^{(N)}$ through *iterative paraphrasing*. The use of iteration is deliberate: a single pass may still retain residual features of the original text, whereas multiple rounds progressively build *human context*. We therefore apply the paraphraser for a fixed number of rounds, producing a sequence $x^{(0)}, x^{(1)}, \dots, x^{(N)}$, where each $x^{(t)}$ rewrites the previous round’s output.

In execution, Algorithm 2 reuses the same prompt structure as training at every round. The current passage $x^{(t-1)}$ is placed into the source field, and the model generates a new target passage $x^{(t)}$. As the number of rounds increases, the semantic content of the text may gradually drift, but the text also moves away from the statistical region occupied by the original generator and toward the paraphraser’s own preferred continuation regime. This trades off semantic retention for humanization.

Algorithm 2 Iterative Paraphrasing

Input: The initial text $x^{(0)}$, the paraphraser $\mathcal{M}_{\text{para}}$, and the number of iterations N

Output: The final paraphrased text $x^{(N)}$

- 1: **for** $t \in \{1, \dots, N\}$ **do**
 - 2: $p^{(t)} \leftarrow \text{FORMATASPROMPT}(x^{(t-1)})$
 - 3: $x^{(t)} \leftarrow \mathcal{M}_{\text{para}}(p^{(t)})$
 - 4: **return** $x^{(N)}$
-

4 Experiments

In this section, we evaluate HIP as a paraphrase-based detector-evasion method across model families, sizes, and baseline methods. We also describe the continuation evaluation introduced in Section 1. We release our code for training and running HIP at <https://github.com/YixuanEvenXu/humanization-by-iterative-paraphrasing>. We release the training and evaluation data, together with the LoRA adapters, through the Hugging Face collection at <https://huggingface.co/collections/YixuanEvenXu/humanization-by-iterative-paraphrasing>.

4.1 Experimental Setup

Datasets. Our experiments require both human-written texts and AI-generated texts from the same domains. We use selected subsets of RAID [Dugan et al., 2024] and MAGE [Li et al., 2024], targeting clean, document-style prose. From RAID, we keep the domains abstracts, books, news, and wiki. From MAGE, we keep the human source families xsum_human, cnn_human, tldr_human, and squad_human, together with their AI counterparts xsum, cnn, tldr, and squad.

- **Training set.** We construct the paired dataset as described in Algorithm 1, with $\mathcal{C}_{\text{raw}}$ being the selected human corpus. After filtering, deduplication, and text-quality screening, each remaining human passage h_i is paraphrased by GPT-5-nano into a_i to form the supervised dataset $\mathcal{D} = \{(a_i, h_i)\}_{i=1}^M$. This process yields a dataset of 11757 training pairs.
- **Evaluation set.** The main evaluation set consists of 256 AI-generated passages, constructed by taking the first 32 examples from each of the eight retained RAID and MAGE source categories.

Evaluation metrics. When evaluating a paraphrasing model or method on the evaluation set, we report three primary metrics. The first is semantic preservation, scored by GPT-5-nano on an integer scale from 0 to 10 by comparing each rewritten text against its original input. A score of 10 denotes complete preservation of meaning, while lower scores indicate greater semantic drift. The other two are detector-based human-likeness scores from the commercial systems GPTZero [Adam et al., 2026] and Pangram [Emi and Spero, 2024]. Both detectors return probability distributions over authorship labels, and we report the probability assigned to the *human* label. Higher values on both detector metrics therefore indicate that a rewritten text is judged more human-like. For qualitative examples that illustrate how these metrics align with actual outputs, see Section B.

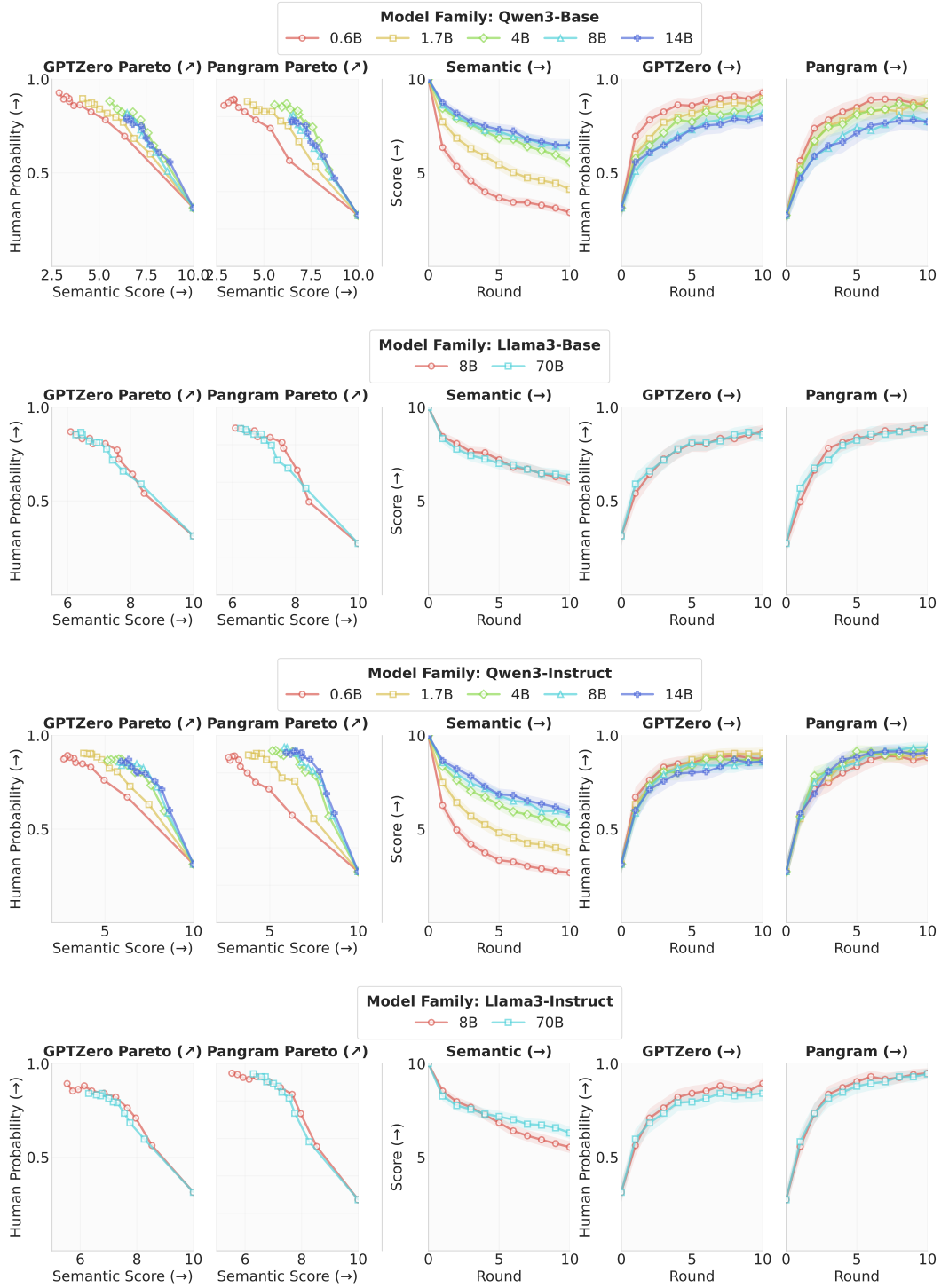


Figure 3: HIP across model families. From top to bottom: Qwen3-Base, Llama3-Base, Qwen3-Instruct, and Llama3-Instruct. Within each panel, the first two subplots show the GPTZero and Pangram Pareto frontiers for the trade-off between semantic preservation and human-likeness, while the last three show semantic score, GPTZero human probability, and Pangram human probability over the rounds, respectively. Shaded areas indicate 95% confidence intervals.